

Week 2 — Command Anatomy

AiShell — more commands: aicd, ails, aipwd, aisysinfo

Summary

- aysinfo: System metrics utility with standard and JSON formatting. Updated original sysinfo to comply with APPANATOMY.
- aipwd: Current directory reporting with logical (-L) and physical (-P) path resolution.
- aicd: Directory navigation supporting tilde expansion, \$HOME, \$OLDPWD (-), and environment synchronization.
- ails: Directory listing supporting combined short flags like -al, long formats, and JSON outputs.
- Infrastructure: Static argtable3 integration, standardized cmd_spec_t module definitions, uniform <name>_run signatures, explicit register_<name>_command() registration, and an automated bash test suite.

Build System & Tooling Enhancements

- POSIX Macro Conformance: Updated C source files to explicitly include `#define _POSIX_C_SOURCE 200809L` prior to system headers, resolving strict `-std=c99 -pedantic` compilation issues with `PATH_MAX` and POSIX glibc extensions.
- Makefile Integration: Updated `CMD_SRCS` and `ALL_BINS` targets to build both individual standalone binaries (`bin/aipwd`, `bin/aicd`, `bin/ails`) and the integrated `bin/aishell` REPL binary.
- Integration Test Framework (`scripts/test.sh`):
 - Migrated inline Makefile test rules to an automated Bash integration test runner.
 - Implemented structured test runners (`run_pass_test` and `run_fail_test`) to validate both successful executions (exit code 0) and error path/invalid flag handling (non-zero exit codes).

Interactive comparison of command flags, positional arguments, and structured JSON output.

Total Commands
4 Core

Global Flags
-h, -v, --json

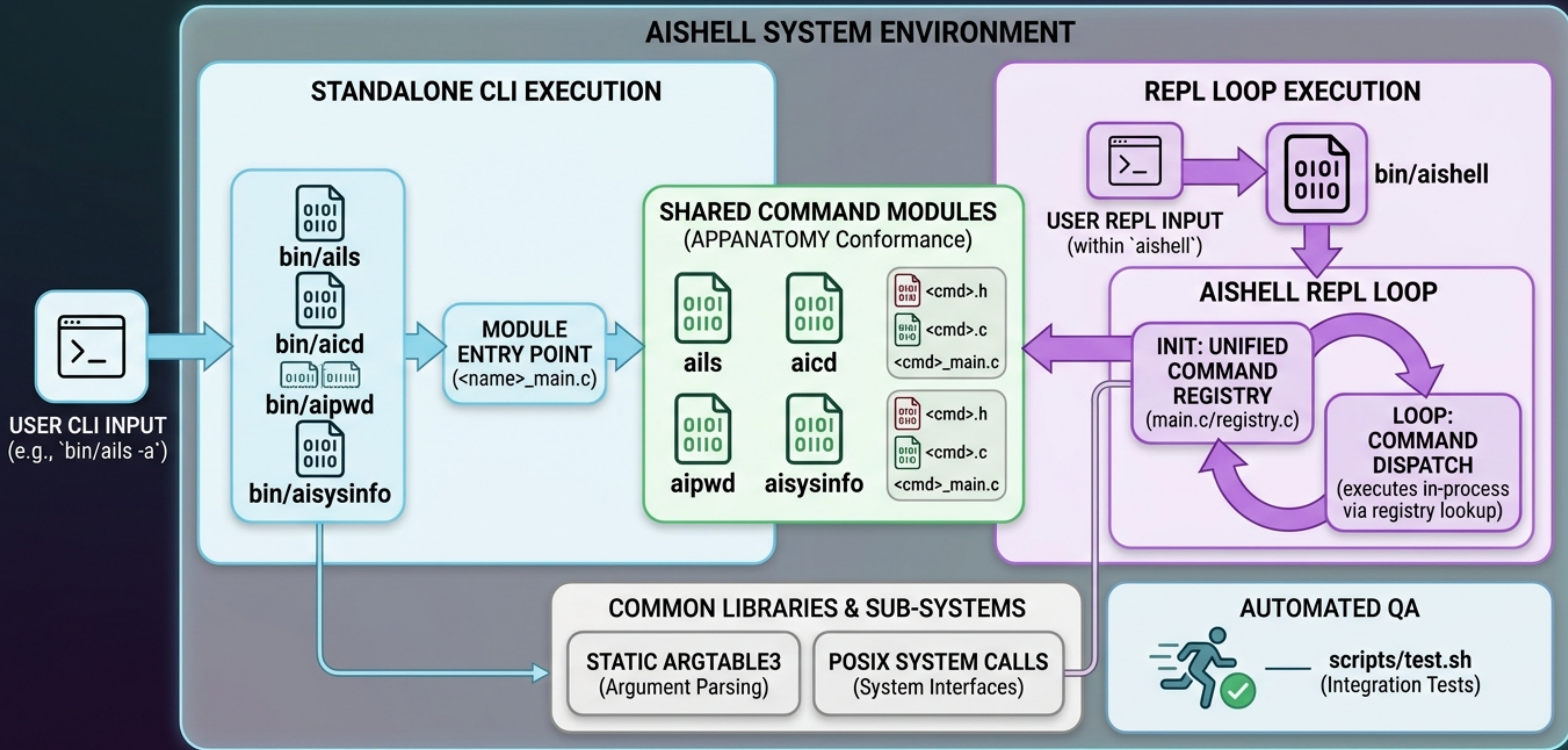
Output Format
JSON / Text

Option Matrix

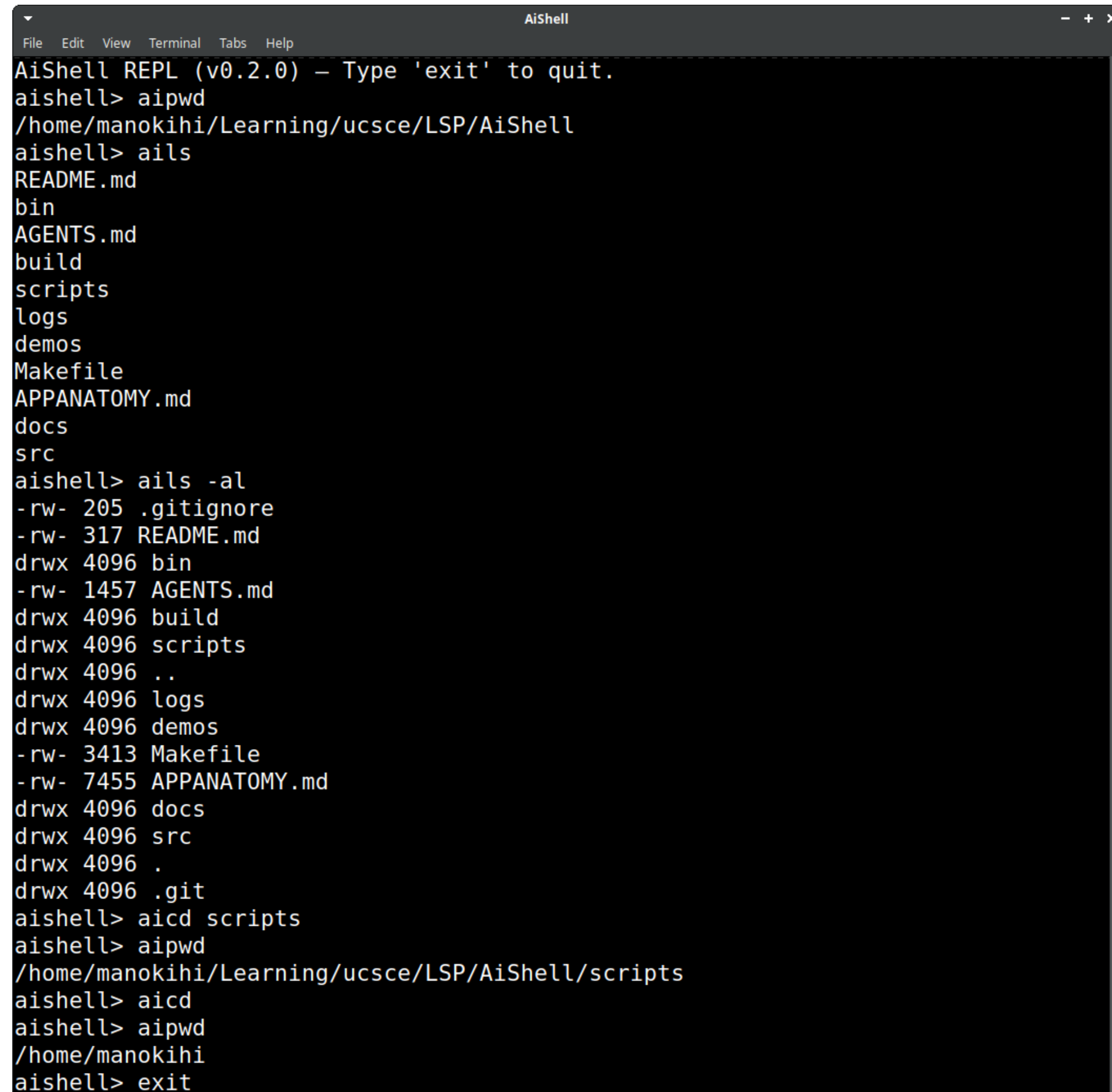
JSON Output Preview

Command	Category	Specific Flags	Positional Args	Global Flags
aicd	Navigation	-L / --logical -P / --physical	[dir]	-h/--help -v/--version --json
aipwd	Navigation	-L / --logical -P / --physical	none	-h/--help -v/--version --json
ails	File System	-a / --all -l / --long	[path]	-h/--help -v/--version --json
aisysinfo	System	-a -s -k -u -m	none	-h/--help -v/--version --json

AIHELL ARCHITECTURAL OVERVIEW: STANDALONE & REPL EXECUTION PATHWAYS



Execution within the AiShell



```
AiShell
File Edit View Terminal Tabs Help
AiShell REPL (v0.2.0) – Type 'exit' to quit.
aishell> aipwd
/home/manokihi/Learning/ucsce/LSP/AiShell
aishell> ails
README.md
bin
AGENTS.md
build
scripts
logs
demos
Makefile
APPANATOMY.md
docs
src
aishell> ails -al
-rw- 205 .gitignore
-rw- 317 README.md
drwx 4096 bin
-rw- 1457 AGENTS.md
drwx 4096 build
drwx 4096 scripts
drwx 4096 ..
drwx 4096 logs
drwx 4096 demos
-rw- 3413 Makefile
-rw- 7455 APPANATOMY.md
drwx 4096 docs
drwx 4096 src
drwx 4096 .
drwx 4096 .git
aishell> aicd scripts
aishell> aipwd
/home/manokihi/Learning/ucsce/LSP/AiShell/scripts
aishell> aicd
aishell> aipwd
/home/manokihi
aishell> exit
```

Tests

```
Test run
File Edit View Terminal Tabs Help
manokihi@mano-ubi:~/Learning/ucsce/LSP/AiShell$ make test
Running tests...
=====
Running AiShell Test Suite
=====
--- aysysinfo ---
Testing [aysysinfo default]... PASS
Testing [aysysinfo --version]... PASS
Testing [aysysinfo --json]... PASS
Testing [aysysinfo -h]... PASS
Testing Error Path [aysysinfo invalid flag]... PASS
Testing Error Path [aysysinfo extra arg]... PASS
--- aipwd ---
Testing [aipwd default]... PASS
Testing [aipwd -L]... PASS
Testing [aipwd -P]... PASS
Testing [aipwd --json]... PASS
Testing [aipwd --version]... PASS
Testing [aipwd --help]... PASS
Testing Error Path [aipwd invalid flag]... PASS
Testing Error Path [aipwd extra positional]... PASS
--- aicd ---
Testing [aicd default (/home/manokihi)]... PASS
Testing [aicd to /tmp]... PASS
Testing [aicd --version]... PASS
Testing [aicd --help]... PASS
Testing [aicd --json]... PASS
Testing Error Path [aicd non-existent dir]... PASS
Testing Error Path [aicd invalid flag]... PASS
Results: 21 Passed, 0 Failed
=====
--- ails ---
Testing [ails default]... PASS
Testing [ails -a]... PASS
Testing [ails -l]... PASS
Testing [ails --version]... PASS
Testing [ails --help]... PASS
Testing [ails --json]... PASS
Testing Error Path [ails non-existent dir]... PASS
Testing Error Path [ails invalid flag]... PASS
--- aishell REPL ---
Testing [aishell aysysinfo]... PASS
Testing [aishell aipwd]... PASS
Testing [aishell aicd]... PASS
Testing [aishell ails]... PASS
Testing Error Path [aishell unknown cmd]... PASS
manokihi@mano-ubi:~/Learning/ucsce/LSP/AiShell$
```